

1. Questions

Study the following data carefully and answer the questions:

Three boats, A, B, and C, are sailing in a river, and the speed of each boat (in m/s) is different.

The difference between the speeds of boats A and B in still water is 4 m/s, and the difference between the speeds of boats B and C in still water is 6 m/s. The time taken by boat B to go 4.2 km in a downstream direction is 3.5 minutes, and the time taken by boat B to go 3.6 km in an upstream direction is 3.75 minutes. The difference between the downstream distance covered by boat A in 85 seconds and that covered by boat C in 105 seconds is 110 m.

What would have been the difference between the upstream distance covered by boat B in 75 seconds and the downstream distance covered by boat C in 50 seconds?

- a. 300 m
- b. 500 m
- c. 150 m
- d. 450 m
- e. 600 m

2. Questions

The difference between the speeds of boats B and D in still water is 3 m/s, and the difference between the speeds of boats C and D in still water is 3 m/s. A man has to cover 690 m in upstream direction. He covered some distance by boat A in 25 seconds and the remaining distance by boat D in 't' seconds. Find the value of 't'?

- a. 50 seconds
- b. 20 seconds
- c. None of these
- d. 40 seconds
- e. 30 seconds

3. Questions

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Three statements, I, II, and III, are given below:

I: The upstream speed of boat A is P% of the downstream speed of boat B.

II: The average of the upstream speed of boat B and the downstream speed of boat C is Q m/s.

III: The downstream speed of boat C is R% more than the upstream speed of boat C.

Find which of the following relations between P, Q, and R is true.

- a. $P = Q + R$

- b. $10P = QR$
- c. $Q^2 = PR$
- d. Both (a) and (b)
- e. Both (a) and (c)

4. Questions

A man travelled by boat A in a downstream direction for 'x' seconds; further, he travelled by boat C in a downstream direction for $(x + 40)$ seconds and covered a total distance of 1.61 km. If he comes back to the starting point by boat C in an upstream direction, then find out in what time he will come back.

- a. Can't be determined
- b. $(5x - 4)$ seconds
- c. $(4x + 21)$ seconds
- d. $(2x + 121)$ seconds
- e. $(6x - 59)$ seconds

5. Questions

The upstream speed of boat E is 12.5% more than the downstream speed of boat A. Boat E travelled for 25 seconds in a downstream direction. If the speed of boat E in still water is decreased by 35% while coming back, then in what time will boat E come back? (Stream speed is always same)

- a. 55 seconds
- b. 25 seconds
- c. 44 seconds
- d. 50 seconds
- e. 40 seconds

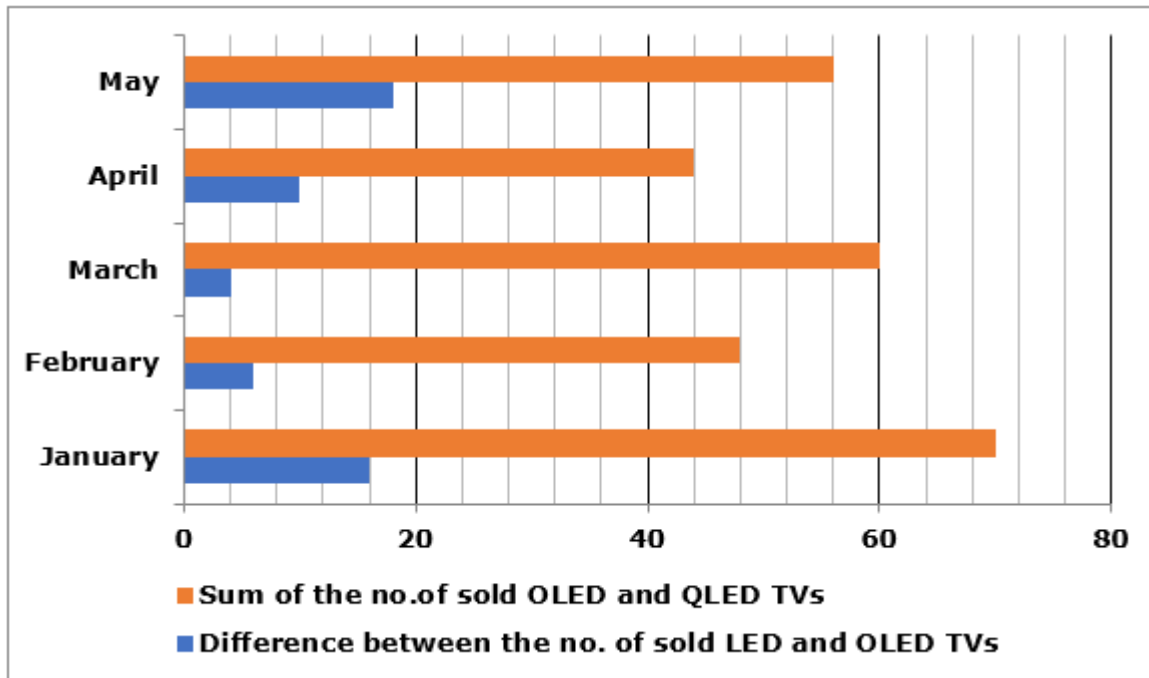
6. Questions

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Study the following data carefully and answer the questions:

The data given below is related to the number of LED TVs, OLED TVs, and QLED TVs sold by a shopkeeper in five months: January, February, March, April, and May.

The bar graph given below shows the following data:



Notes:

- 1). The ratio of the number of LED TVs sold in January to that sold in April is 9: 5, the number of LED TVs sold in January is a multiple of 5 and the number of LED TVs sold in April is not more than 30.
- 2). The average of the number of LED TVs sold in January, April and May is 30.
- 3). The numbers of LED TVs sold in February and March are respectively 140% and 175% of that sold in May.

In February, if the number of LED TVs sold were more than the number of OLED TVs sold, then the number of QLED TVs sold would have been M, and if the number of LED TVs sold were less than the number of OLED TVs sold, then the number of QLED TVs sold would have been N. Find the ratio of M to N?

- a. None of these
- b. 13: 7
- c. 25: 14
- d. 11: 5
- e. 20: 11

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7. Questions

In April, if the price of a LED TV is ₹9600 per unit, the price of a OLED TV is ₹12000 per unit, the price of a QLED TV is ₹10000 per unit, and the total amount received by the shopkeeper by selling LED and OLED TVs together in April is not a multiple of 7, then find the amount received by the shopkeeper by selling QLED TVs in April.

- a. ₹290000
- b. ₹440000

- c. ₹340000
- d. ₹90000
- e. ₹180000

8. Questions

If in January and May the number of sold LED TVs is less than the number of sold OLED TVs, then what will be the ratio of the number of QLED TVs sold in January to that sold in May?

- a. 3: 5
- b. 1: 2
- c. 1: 1
- d. 9: 17
- e. 2: 3

9. Questions

If the number of OLED TVs sold in March is a multiple of 3 and the number of OLED TVs sold in April is also a multiple of 3, then find which of the following statements is/are true:?

P: The ratio of the number of QLED TVs sold in March to the number of OLED TVs sold in April is 7: 5.

Q: The average of the number of OLED TVs sold in March and the number of QLED TVs sold in April is 34.

R: The sum of the number of QLED TVs sold in March and April is a multiple of 5.

- a. All are true
- b. Only Q and R
- c. Only P
- d. Only P and Q
- e. Only Q

10. Questions

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If the number of LED TVs sold in January is more than the number of OLED TVs sold in January, the average number of LED, OLED, and QLED TVs sold in January and June together is 42, 31, and 31, respectively. Then find the average of the number of LED, OLED, and QLED TVs sold in June.

- a. 39
- b. 27
- c. 31
- d. 36

e. 33

11. Questions

In each of the following questions, read the given statement and compare Quantity I, Quantity II and Quantity III on its basis. (Only quantity is to be considered)

The simple interest received on ₹15000 at an 8% rate after T years is equal to the simple interest received on ₹18000 at an R% rate after 4 years, and the simple interest received on ₹25000 at an R% rate after 2T years is ₹7500.

Quantity I: Value of T is what percent of R?

Quantity II: Find the value of $(T^3 + R^2)$.

Quantity III: Find the difference between the simple interest and compound interest on ₹16000 at R% rate after 2 years.

- a. Quantity I < Quantity II < Quantity III
- b. Quantity I > Quantity II > Quantity III
- c. Quantity I > Quantity II = Quantity III
- d. Quantity I = Quantity II < Quantity III
- e. Quantity I = Quantity II = Quantity III

12. Questions

In a mixture of milk and water, the quantity of milk is 48 L and the cost of pure milk is ₹75 per L. The mixture is sold at ₹81 per L and 35% profit is earned.

Quantity I: Find the difference between the quantities of milk and water in the mixture.

Quantity II: If the cost of pure milk were ₹50 per L, then what would have been the cost of the mixture?

Quantity III: If $16\frac{2}{3}\%$ of the quantity of the mixture is replaced with the same quantity of water, then what will be the new quantity of milk in the mixture?

- a. Quantity I < Quantity II < Quantity III
- b. Quantity I > Quantity II > Quantity III
- c. Quantity I = Quantity II > Quantity III
- d. Quantity I < Quantity II = Quantity III
- e. Quantity I = Quantity II = Quantity III

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13. Questions

The average of four numbers is 31. The ratio of the smallest number to the 2nd largest number is 1:2, the ratio of the 2nd smallest number to the largest number is 5:9, and 150% of the smallest number is equal to 60% of the largest number.

Quantity I: What is the average of the smallest number, the 2nd largest number, and the largest number?

Quantity II: Find the difference between 50% of the smallest number and $88\frac{8}{9}\%$ of the largest number.

Quantity III: If a new number is 3 less than the smallest number, then find that the new number is what percent of the third largest number?

- a. Quantity I > Quantity II > Quantity III
- b. Quantity I > Quantity III > Quantity II
- c. Quantity III > Quantity I > Quantity II
- d. Quantity II > Quantity I > Quantity III
- e. Quantity III > Quantity II > Quantity I

14. Questions

A shopkeeper sold two articles, A and B, each for ₹7200, and he neither received a profit nor obtained a loss for selling the two articles together. He sold article A at X% profit and sold article B at a $14\frac{2}{7}\%$ loss.

Quantity I: If the shopkeeper had marked up article A by $(X + 5)\%$ and sold it at a ₹900 discount, then what would have been the profit/loss percentage on article A?

Quantity II: If article A were sold at a 10% loss and article B were sold at a 25% profit, then what would have been the total profit/loss percentage for selling two articles together?

Quantity III: The cost price of article B is what percent more than that of article A?

- a. Quantity I > Quantity II > Quantity III
- b. Quantity I > Quantity III > Quantity II
- c. Quantity III > Quantity I > Quantity II
- d. Quantity II > Quantity I > Quantity III
- e. Quantity III > Quantity II > Quantity I

15. Questions

The speed of bike A is 18 km/h less than that of bike B and 10.8 km/h less than that of bike C. The distance covered by bike B in 4.5 minutes is 2.7 km less than that covered by boat C in 7.5 minutes.

Quantity I: If the distance between bikes A and C is 825 m and they are coming towards each other, then in what time they will meet each other?

Quantity II: The speed of bike A is what percent less than that of bike B?

Quantity III: If the speed of bike D is 80% of that of bike B, then find the time taken by bike D to cross a 352 m long tunnel.

- a. Quantity I = Quantity II > Quantity III

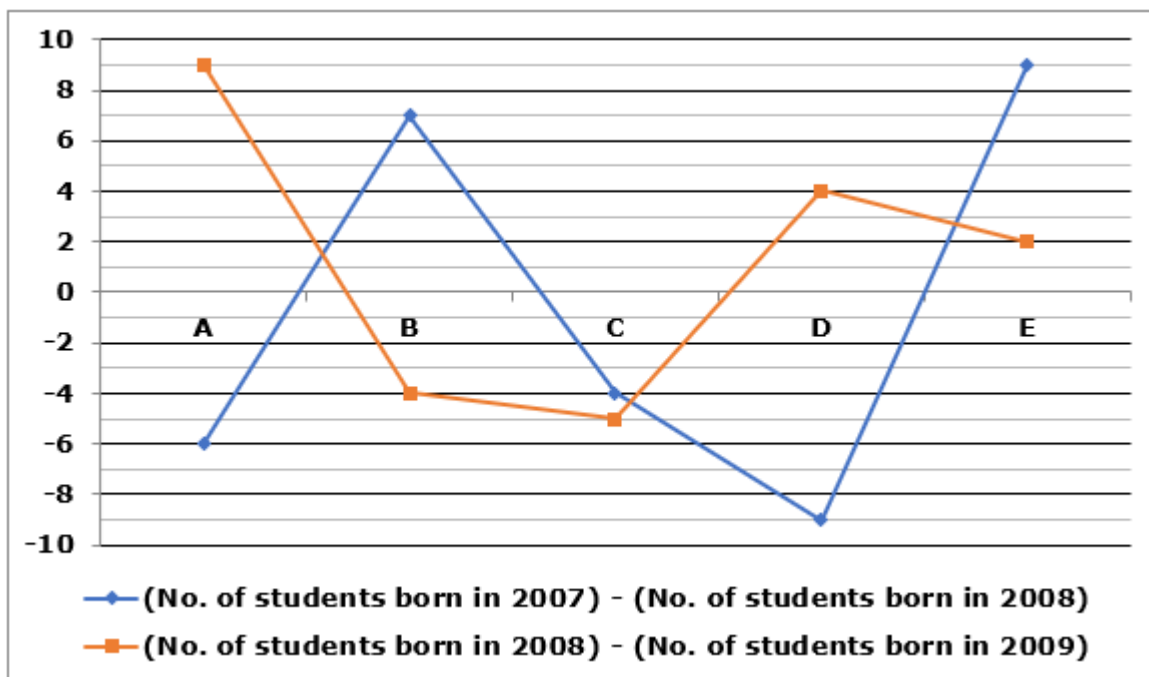
- b. Quantity I > Quantity III = Quantity II
- c. Quantity III > Quantity I > Quantity II
- d. Quantity II = Quantity I = Quantity III
- e. Quantity III > Quantity II > Quantity I

16. Questions

Study the following data carefully and answer the questions:

There are five classes, A, B, C, D, and E, and the number of students in each class is different. The birth year of each student of each class is either 2007 or 2008, or 2009.

The bar graph given below shows the following data:



Notes:

- 1: The number of students in class C, who were born in 2009, is 5 less than double the number of students in class A, who were born in 2009.
- 2: The number of students in classes B and D, who were born in 2009, are prime numbers between 15 and 22 and the number of students in class B, who were born in 2009, is more than those in class D, who were born in 2009.
- 3: The number of students in class A, who were born in 2009, is 2 less than those in class D, who were born in 2009 and also 1 more than those in class E, who were born in 2009.

If the average age of class D in 2022 is M years, then find the average of the ages of those students of class E in 2020, who were born in 2008, and the ages of those students of class E in 2024, who were born in 2009.

- a. $(M - 1.5)$ years
- b. $(M - 0.25)$ years

- c. $(M - 0.5)$ years
- d. $(M - 0.75)$ years
- e. $(M - 0.1)$ years

17. Questions

If 6 students of class B, who were born in 2008, left class B, then what will be the average age of class B in 2021?

- a. 12.46 years
- b. 13.06 years
- c. 11.16 years
- d. 14.56 years
- e. 12.96 years

18. Questions

In class E, if the ratio of boys to girls, who were born in 2007, is 3: 2, the ratio of boys to girls, who were born in 2008, is 3:5, and the ratio of boys to girls, who were born in 2009, is 2: 5, then find the percentage of girls in class E.

- a. $33\frac{1}{3}\%$
- b. $44\frac{4}{9}\%$
- c. 40%
- d. 30%
- e. $54\frac{6}{11}\%$

19. Questions

If the average of the ages of those students of class A in 2022, who were born in 2007 and the ages of those students of class D in 2017, who were born in 2007, is P years. the average of the ages of those students of class A in 2025, who were born in 2009 and the ages of those students of class C in 2025, who were born in 2009, is Q years, then find that which of the following is/are true?

X: P and Q are co-prime with each other.

Y: Value of $(Q^2 - P^2)$ is a prime number.

Z: Value of $(P^2 + Q^2)$ is divisible by 25.

- a. Only X and Z
- b. All are true
- c. Only X and Y

d. Only Y

e. Only Z

20. Questions

If the number of students left in class C, who were born in 2008, were $[10(Q - P)]\%$ of the number of students in class C, who were born in 2008. What is the total number of students in class C now? (Consider the values of P and Q from the above question.)

a. 44

b. 55

c. 45

d. 65

e. 50

Explanations:

1. Questions

Since, the time taken by boat B to go 4.2 km in downstream direction is 3.5 minutes.

So, the downstream speed of boat B = $(4.2 * 1000)/(3.5 * 60) = 20$ m/s

Since, the time taken by boat B to go 3.6 km in upstream direction is 3.75 minutes.

So, the upstream speed of boat B = $(3.6 * 1000)/(3.75 * 60) = 16$ m/s

The speed of boat B in still water = $(20 + 16)/2 = 18$ m/s

And the speed of stream = $(20 - 16)/2 = 2$ m/s

Since, the difference between the speeds of boats A and B in still water is 4 m/s.

So, the speed of boat A in still water will either be $(18 - 4) = 14$ m/s or $(18 + 4) = 22$ m/s

Since, the difference between the speeds of boats B and C in still water is 6 m/s.

So, the speed of boat C in still water will either be $(18 - 6) = 12$ m/s or $(18 + 6) = 24$ m/s

Since, the difference between the downstream distance covered by boat A in 85 seconds and that covered by boat C in 105 seconds is 110 m.

Case I:

When the speed of boat A in still water = 14 m/s

And the speed of boat C in still water = 12 m/s

So, $(14 + 2) * 85 \sim (12 + 2) * 105 = 1360 \sim 1470 = 110$ m

Case II:

When the speed of boat A in still water = 14 m/s

And the speed of boat C in still water = 24 m/s

So, $(14 + 2) * 85 \sim (24 + 2) * 105 = 1360 \sim 2730 = 1370$ m

Case III:

When the speed of boat A in still water = 22 m/s

And the speed of boat C in still water = 12 m/s

So, $(22 + 2) * 85 \sim (12 + 2) * 105 = 2040 \sim 1470 = 570$ m

Case IV:

When the speed of boat A in still water = 22 m/s

And the speed of boat C in still water = 24 m/s

So, $(22 + 2) * 85 \sim (24 + 2) * 105 = 2040 \sim 2730 = 690$ m

Since, cases II, III and IV can be neglected.

So, the speed of boat A in still water = 14 m/s

The speed of boat B in still water = 18 m/s

And the speed of boat C in still water = 12 m/s

Answer: B

The downstream distance covered by boat C in 50 seconds would have been = $(12 + 2) * 50 = 700$ m

The upstream distance covered by boat B in 75 seconds = $(18 - 2) * 75 = 1200$ m

Required difference = $1200 - 700 = 500$ m

2. Questions

Since, the time taken by boat B to go 4.2 km in downstream direction is 3.5 minutes.

So, the downstream speed of boat B = $(4.2 * 1000)/(3.5 * 60) = 20$ m/s

Since, the time taken by boat B to go 3.6 km in upstream direction is 3.75 minutes.

So, the upstream speed of boat B = $(3.6 * 1000)/(3.75 * 60) = 16$ m/s

The speed of boat B in still water = $(20 + 16)/2 = 18$ m/s

And the speed of stream = $(20 - 16)/2 = 2$ m/s

Since, the difference between the speeds of boats A and B in still water is 4 m/s.

So, the speed of boat A in still water will either be $(18 - 4) = 14$ m/s or $(18 + 4) = 22$ m/s

Since, the difference between the speeds of boats B and C in still water is 6 m/s.

So, the speed of boat C in still water will either be $(18 - 6) = 12$ m/s or $(18 + 6) = 24$ m/s

Since, the difference between the downstream distance covered by boat A in 85 seconds and that covered

by boat C in 105 seconds is 110 m.

Case I:

When the speed of boat A in still water = 14 m/s

And the speed of boat C in still water = 12 m/s

So, $(14 + 2) * 85 \sim (12 + 2) * 105 = 1360 \sim 1470 = 110 \text{ m}$

Case II:

When the speed of boat A in still water = 14 m/s

And the speed of boat C in still water = 24 m/s

So, $(14 + 2) * 85 \sim (24 + 2) * 105 = 1360 \sim 2730 = 1370 \text{ m}$

Case III:

When the speed of boat A in still water = 22 m/s

And the speed of boat C in still water = 12 m/s

So, $(22 + 2) * 85 \sim (12 + 2) * 105 = 2040 \sim 1470 = 570 \text{ m}$

Case IV:

When the speed of boat A in still water = 22 m/s

And the speed of boat C in still water = 24 m/s

So, $(22 + 2) * 85 \sim (24 + 2) * 105 = 2040 \sim 2730 = 690 \text{ m}$

Since, cases II, III and IV can be neglected.

So, the speed of boat A in still water = 14 m/s

The speed of boat B in still water = 18 m/s

And the speed of boat C in still water = 12 m/s

Answer: E

Since, the difference between the speeds of boats B and D in still water is 3 m/s and the difference between the speeds of boats C and D in still water is 3 m/s.

So, the only possible value of the speed of boat D in still water = $(18 - 3)$ or $(12 + 3) = 15 \text{ m/s}$

Since, the speed of boat A in still water = 14 m/s

And the speed of stream = 2 m/s

So, the upstream speed of boat A = $(14 - 2) = 12 \text{ m/s}$

And the upstream speed of boat D = $(15 - 2) = 13 \text{ m/s}$

Now,

$(12 * 25) + (13 * t) = 690$

$t = 30$ seconds

3. Questions

Since, the time taken by boat B to go 4.2 km in downstream direction is 3.5 minutes.

So, the downstream speed of boat B = $(4.2 * 1000)/(3.5 * 60) = 20$ m/s

Since, the time taken by boat B to go 3.6 km in upstream direction is 3.75 minutes.

So, the upstream speed of boat B = $(3.6 * 1000)/(3.75 * 60) = 16$ m/s

The speed of boat B in still water = $(20 + 16)/2 = 18$ m/s

And the speed of stream = $(20 - 16)/2 = 2$ m/s

Since, the difference between the speeds of boats A and B in still water is 4 m/s.

So, the speed of boat A in still water will either be $(18 - 4) = 14$ m/s or $(18 + 4) = 22$ m/s

Since, the difference between the speeds of boats B and C in still water is 6 m/s.

So, the speed of boat C in still water will either be $(18 - 6) = 12$ m/s or $(18 + 6) = 24$ m/s

Since, the difference between the downstream distance covered by boat A in 85 seconds and that covered by boat C in 105 seconds is 110 m.

Case I:

When the speed of boat A in still water = 14 m/s

And the speed of boat C in still water = 12 m/s

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Case II:

When the speed of boat A in still water = 14 m/s

And the speed of boat C in still water = 24 m/s

So, $(14 + 2) * 85 \sim (24 + 2) * 105 = 1360 \sim 2730 = 1370$ m

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When the speed of boat A in still water = 22 m/s

And the speed of boat C in still water = 12 m/s

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Case IV:

When the speed of boat A in still water = 22 m/s

And the speed of boat C in still water = 24 m/s

So, $(22 + 2) * 85 \sim (24 + 2) * 105 = 2040 \sim 2730 = 690$ m

Since, cases II, III and IV can be neglected.

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So, the speed of boat A in still water = 14 m/s

The speed of boat B in still water = 18 m/s

And the speed of boat C in still water = 12 m/s

Answer: B

From statement I:

Since, the upstream speed of boat A = $14 - 2 = 12$ m/s

And the downstream speed of boat B = $18 + 2 = 20$ m/s

So, $P = (12/20) * 100$

$P = 60\%$

From statement II:

Since, the upstream speed of boat B = $18 - 2 = 16$ m/s

And the downstream speed of boat C = $12 + 2 = 14$ m/s

So, $Q = (16 + 14)/2$

$Q = 15$ m/s

From statement III:

Since, the downstream speed of boat C = $12 + 2 = 14$ m/s

And the upstream speed of boat C = $12 - 2 = 10$ m/s

So, $R = [(14 - 10)/10] * 100$

$R = 40\%$

From option (a):

$P = Q + R$

$60 = 15 + 40$

$60 = 55$

So, option (a) is not correct.

From option (b):

$10P = QR$

$10 * 60 = 15 * 40$

$600 = 600$

So, option (b) is correct.

From option (c):

$Q^2 = PR$

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$$15^2 = 60 * 40$$

$$225 = 240$$

So, option (c) is not correct.

Hence, option (b) is correct.

4. Questions

Since, the time taken by boat B to go 4.2 km in downstream direction is 3.5 minutes.

So, the downstream speed of boat B = $(4.2 * 1000)/(3.5 * 60) = 20$ m/s

Since, the time taken by boat B to go 3.6 km in upstream direction is 3.75 minutes.

So, the upstream speed of boat B = $(3.6 * 1000)/(3.75 * 60) = 16$ m/s

The speed of boat B in still water = $(20 + 16)/2 = 18$ m/s

And the speed of stream = $(20 - 16)/2 = 2$ m/s

Since, the difference between the speeds of boats A and B in still water is 4 m/s.

So, the speed of boat A in still water will either be $(18 - 4) = 14$ m/s or $(18 + 4) = 22$ m/s

Since, the difference between the speeds of boats B and C in still water is 6 m/s.

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Since, the difference between the downstream distance covered by boat A in 85 seconds and that covered by boat C in 105 seconds is 110 m.

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So, $(14 + 2) * 85 \sim (12 + 2) * 105 = 1360 \sim 1470 = 110$ m

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When the speed of boat A in still water = 14 m/s

And the speed of boat C in still water = 24 m/s

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Case III:

When the speed of boat A in still water = 22 m/s

And the speed of boat C in still water = 12 m/s

So, $(22 + 2) * 85 \sim (12 + 2) * 105 = 2040 \sim 1470 = 570$ m

Case IV:

When the speed of boat A in still water = 22 m/s

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And the speed of boat C in still water = 24 m/s

So, $(22 + 2) * 85 \sim (24 + 2) * 105 = 2040 \sim 2730 = 690$ m

Since, cases II, III and IV can be neglected.

So, the speed of boat A in still water = 14 m/s

The speed of boat B in still water = 18 m/s

And the speed of boat C in still water = 12 m/s

Answer: C

Since, the downstream speed of boat A = $14 + 2 = 16$ m/s

And the downstream speed of boat C = $12 + 2 = 14$ m/s

So,

$$[16 * x] + [14 * (x + 40)] = 1.61 * 1000$$

$$16x + 14x + 560 = 1610$$

$$30x = 1050$$

$$x = 35$$

Since, the upstream speed of boat C = $12 - 2 = 10$ m/s

So, the time, in which he will come back = $(1.61 * 1000)/10 = 161$ seconds = $(4x + 21)$ seconds

5. Questions

Since, the time taken by boat B to go 4.2 km in downstream direction is 3.5 minutes.

So, the downstream speed of boat B = $(4.2 * 1000)/(3.5 * 60) = 20$ m/s

Since, the time taken by boat B to go 3.6 km in upstream direction is 3.75 minutes.

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So, the speed of boat A in still water will either be $(18 - 4) = 14$ m/s or $(18 + 4) = 22$ m/s

Since, the difference between the speeds of boats B and C in still water is 6 m/s.

So, the speed of boat C in still water will either be $(18 - 6) = 12$ m/s or $(18 + 6) = 24$ m/s

Since, the difference between the downstream distance covered by boat A in 85 seconds and that covered by boat C in 105 seconds is 110 m.

Case I:

When the speed of boat A in still water = 14 m/s

And the speed of boat C in still water = 12 m/s

So, $(14 + 2) * 85 \sim (12 + 2) * 105 = 1360 \sim 1470 = 110$ m

Case II:

When the speed of boat A in still water = 14 m/s

And the speed of boat C in still water = 24 m/s

So, $(14 + 2) * 85 \sim (24 + 2) * 105 = 1360 \sim 2730 = 1370$ m

Case III:

When the speed of boat A in still water = 22 m/s

And the speed of boat C in still water = 12 m/s

So, $(22 + 2) * 85 \sim (12 + 2) * 105 = 2040 \sim 1470 = 570$ m

Case IV:

When the speed of boat A in still water = 22 m/s

And the speed of boat C in still water = 24 m/s

So, $(22 + 2) * 85 \sim (24 + 2) * 105 = 2040 \sim 2730 = 690$ m

Since, cases II, III and IV can be neglected.

So, the speed of boat A in still water = 14 m/s

The speed of boat B in still water = 18 m/s

And the speed of boat C in still water = 12 m/s

Answer: D

Since, the downstream speed of boat A = $14 + 2 = 16$ m/s

So, the upstream speed of boat E = 112.5% of 16 = 18 m/s

And the speed of boat E in still water = $18 + 2 = 20$ m/s

Since, the downstream speed of boat E = $20 + 2 = 22$ m/s

So, the downstream distance covered by boat E = $(20 + 2) * 25 = 550$ m

Since, the speed of boat E in still water is decreased by 35% while coming back.

So, the upstream speed of boat E while coming back = 65% of $20 - 2 = 11$ m/s

And the time taken by boat E to come back = $550/11 = 50$ seconds

6. Questions

Since, the ratio of the number of LED TVs sold in January to that sold in April is 9: 5 and the number of LED TVs sold in January is a multiple of 5.

So, the number of LED TVs sold in January will be a multiple of $(9 * 5) = 45$

If the number of LED TVs sold in January = 45

So, the number of LED TVs sold in April = $45 * (5/9) = 25$

If we took the value of the number of LED TVs sold in January more than 45:

Then the number of LED TVs sold in April will be more than 30.

So, the number of LED TVs sold in January = 45

And the number of LED TVs sold in April = 25

Since, the average of the number of LED TVs sold in January, April and May is 30.

So, the number of LED TVs sold in May = $3 * 30 - (45 + 25) = 20$

Since, the number of LED TVs sold in February is 140% of that sold in May.

So, the number of LED TVs sold in February = $140\% \text{ of } 20 = 28$

Since, the number of LED TVs sold in March is 175% of that sold in May.

So, the number of LED TVs sold in March = $175\% \text{ of } 20 = 35$

Month	No. of sold LED TVs	Difference between the no. of sold LED and OLED TVs	Sum of the no. of sold OLED and QLED TVs
January	45	16	70
February	28	6	48
March	35	4	60
April	25	10	44
May	20	18	56

Answer: B

In February:

The number of sold LED TVs = 28

If the number of sold LED TVs were more than the number of sold OLED TVs:

Then the number of sold OLED TVs would have been = $28 - 6 = 22$ @QuizByMickey

And the number of sold QLED TVs would have been = $M = 48 - 22 = 26$

If the number of sold LED TVs were less than the number of sold OLED TVs:

Then the number of sold OLED TVs would have been = $28 + 6 = 34$

And the number of sold QLED TVs would have been = $N = 48 - 34 = 14$

So, the required ratio = $M : N = 26 : 14 = 13 : 7$

7. Questions

Since, the ratio of the number of LED TVs sold in January to that sold in April is 9: 5 and the number of LED TVs sold in January is a multiple of 5.

So, the number of LED TVs sold in January will be a multiple of $(9 * 5) = 45$

If the number of LED TVs sold in January = 45

So, the number of LED TVs sold in April = $45 * (5/9) = 25$

If we took the value of the number of LED TVs sold in January more than 45:

Then the number of LED TVs sold in April will be more than 30.

So, the number of LED TVs sold in January = 45

And the number of LED TVs sold in April = 25

Since, the average of the number of LED TVs sold in January, April and May is 30.

So, the number of LED TVs sold in May = $3 * 30 - (45 + 25) = 20$

Since, the number of LED TVs sold in February is 140% of that sold in May.

So, the number of LED TVs sold in February = 140% of 20 = 28

Since, the number of LED TVs sold in March is 175% of that sold in May.

So, the number of LED TVs sold in March = 175% of 20 = 35

Month	No. of sold LED TVs	Difference between the no. of sold LED and OLED TVs	Sum of the no. of sold OLED and QLED TVs
January	45	16	70
February	28	6	48
March	35	4	60
April	25	10	44
May	20	18	56

Answer: D

Since, the number of LED TVs sold in April = 25

So, the number of OLED TVs sold in April will either be $25 + 10 = 35$ or $25 - 10 = 15$

Case I: If the number of OLED TVs sold in April is 35:

Then the total amount received by selling LED and OLED TVs together in April = $(25 * 9600) + (35 * 12000) = 240000 + 420000 = ₹660000$

Case II: If the number of OLED TVs sold in April is 15:

Then the total amount received by selling LED and OLED TVs together in April = $(25 * 9600) + (15 * 12000) = 240000 + 180000 = ₹420000$

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Since, the total amount received by selling LED and OLED TVs together in April is not a multiple of 7.

So, case II can be neglected.

Since, the number of OLED TVs sold in April = 35

So, the number of QLED TVs sold in April = $44 - 35 = 9$

And the total amount received by selling QLED TVs in April = $9 * 10000 = ₹90000$

8. Questions

Since, the ratio of the number of LED TVs sold in January to that sold in April is 9: 5 and the number of LED TVs sold in January is a multiple of 5.

So, the number of LED TVs sold in January will be a multiple of $(9 * 5) = 45$

If the number of LED TVs sold in January = 45

So, the number of LED TVs sold in April = $45 * (5/9) = 25$

If we took the value of the number of LED TVs sold in January more than 45:

Then the number of LED TVs sold in April will be more than 30.

So, the number of LED TVs sold in January = 45

And the number of LED TVs sold in April = 25

Since, the average of the number of LED TVs sold in January, April and May is 30.

So, the number of LED TVs sold in May = $3 * 30 - (45 + 25) = 20$

Since, the number of LED TVs sold in February is 140% of that sold in May.

So, the number of LED TVs sold in February = $140\% \text{ of } 20 = 28$

Since, the number of LED TVs sold in March is 175% of that sold in May.

So, the number of LED TVs sold in March = $175\% \text{ of } 20 = 35$

Month	No. of sold LED TVs	Difference between the no. of sold LED and OLED TVs	Sum of the no. of sold OLED and QLED TVs
January	45	16	70
February	28	6	48
March	35	4	60
April	25	10	44
May	20	18	56

Answer: B

In January:

Since, the number of sold LED TVs = 45

And the number of sold LED TVs is less than the number of sold OLED TVs.

So, the number of sold OLED TVs = $45 + 16 = 61$

And the number of sold QLED TVs = $70 - 61 = 9$

In May:

Since, the number of sold LED TVs = 20

And the number of sold LED TVs is less than the number of sold OLED TVs.

So, the number of sold OLED TVs = $20 + 18 = 38$

And the number of sold QLED TVs = $56 - 38 = 18$

Required ratio = $9:18 = 1:2$

9. Questions

Since, the ratio of the number of LED TVs sold in January to that sold in April is 9: 5 and the number of LED TVs sold in January is a multiple of 5.

So, the number of LED TVs sold in January will be a multiple of $(9 * 5) = 45$

If the number of LED TVs sold in January = 45

So, the number of LED TVs sold in April = $45 * (5/9) = 25$

If we took the value of the number of LED TVs sold in January more than 45:

Then the number of LED TVs sold in April will be more than 30.

So, the number of LED TVs sold in January = 45

And the number of LED TVs sold in April = 25

Since, the average of the number of LED TVs sold in January, April and May is 30.

So, the number of LED TVs sold in May = $3 * 30 - (45 + 25) = 20$

Since, the number of LED TVs sold in February is 140% of that sold in May.

So, the number of LED TVs sold in February = $140\% \text{ of } 20 = 28$

Since, the number of LED TVs sold in March is 175% of that sold in May.

So, the number of LED TVs sold in March = $175\% \text{ of } 20 = 35$

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Month	No. of sold LED TVs	Difference between the no. of sold LED and OLED TVs	Sum of the no. of sold OLED and QLED TVs
January	45	16	70
February	28	6	48
March	35	4	60
April	25	10	44
May	20	18	56

Answer: A

Since, the number of LED TVs sold in March = 35

So, the number of OLED TVs sold in March will either be $35 + 4 = 39$ or $35 - 4 = 31$.

Since, the number of OLED TVs sold in March is a multiple of 3.

So, the number of OLED TVs sold in March = 39

And the number of QLED TVs sold in March = $60 - 39 = 21$

Since, the number of LED TVs sold in April = 25

So, the number of OLED TVs sold in April will either be $25 + 10 = 35$ or $25 - 10 = 15$

Since, the number of OLED TVs sold in April is a multiple of 3.

So, the number of OLED TVs sold in April = 15

And the number of QLED TVs sold in April = $44 - 15 = 29$

From P:

The number of QLED TVs sold in March = 21

The number of OLED TVs sold in April = 15

The required ratio = $21:15 = 7:5$

So, P is true.

From Q:

The number of OLED TVs sold in March = 39

The number of QLED TVs sold in April = 29

The required average = $(39 + 29)/2 = 34$

So, Q is true.

From R:

The number of QLED TVs sold in March = 21

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The number of QLED TVs sold in April = 29

The required sum = $21 + 29 = 50$

Since, 50 is a multiple of 5.

So, R is true.

Hence, all are true.

10. Questions

Since, the ratio of the number of LED TVs sold in January to that sold in April is 9: 5 and the number of LED TVs sold in January is a multiple of 5.

So, the number of LED TVs sold in January will be a multiple of $(9 * 5) = 45$

If the number of LED TVs sold in January = 45

So, the number of LED TVs sold in April = $45 * (5/9) = 25$

If we took the value of the number of LED TVs sold in January more than 45:

Then the number of LED TVs sold in April will be more than 30.

So, the number of LED TVs sold in January = 45

And the number of LED TVs sold in April = 25

Since, the average of the number of LED TVs sold in January, April and May is 30.

So, the number of LED TVs sold in May = $3 * 30 - (45 + 25) = 20$

Since, the number of LED TVs sold in February is 140% of that sold in May.

So, the number of LED TVs sold in February = $140\% \text{ of } 20 = 28$

Since, the number of LED TVs sold in March is 175% of that sold in May.

So, the number of LED TVs sold in March = $175\% \text{ of } 20 = 35$

Month	No. of sold LED TVs	Difference between the no. of sold LED and OLED TVs	Sum of the no. of sold OLED and QLED TVs
January	45	16	70
February	28	6	48
March	35	4	60
April	25	10	44
May	20	18	56

Answer: C

Since, the number of LED TVs sold in January = 45

And the number of LED TVs sold in January is more than the number of OLED TVs sold in January.

So, the number of OLED TVs sold in January = $45 - 16 = 29$

And the number of QLED TVs sold in January = $70 - 29 = 41$

Since, the average number of LED TVs sold in January and June = 42

So, the number of LED TVs sold in June = $(2 * 42) - 45 = 39$

Since, the average number of OLED TVs sold in January and June = 31

So, the number of OLED TVs sold in June = $(2 * 31) - 29 = 33$

Since, the average number of QLED TVs sold in January and June = 31

So, the number of QLED TVs sold in June = $(2 * 31) - 41 = 21$

Required average = $(39 + 33 + 21)/3 = 31$

11. Questions

Answer: B

Since, the SI received on ₹15000 at 8% rate after T years is equal to the simple interest received on ₹18000 at R% rate after 4 years.

So,

$$(15000 * 8 * T)/100 = (18000 * R * 4)/100$$

$$15 * 8 * T = 18 * R * 4$$

$$5T = 3R$$

$$T = 3R/5 \text{ -----(1)}$$

Since, the SI received on ₹25000 at R% rate after 2T years is ₹7500.

So,

$$(25000 * R * 2T)/100 = 7500$$

$$RT = 15 \text{ -----(2)}$$

From equations (1) and (2):

$$R * (3R/5) = 15$$

$$R^2 = 25$$

$$R = 5$$

From equation (1):

$$T = (3 * 5)/5$$

$$T = 3$$

Quantity I:

Since, T = 3

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And, $R = 5$

So, the required percentage $= (3/5) * 100 = 60\%$

Quantity II:

Since, $T = 3$

And, $R = 5$

So, the value of $(T^3 + R^2) = (3^3 + 5^2) = 27 + 25 = 52$

Quantity III:

The difference between CI and SI on ₹16000 at $R\%$ rate after 2 years $= 16000 * (R/100)^2$

$= 16000 * (5/100)^2$

$= ₹40$

Hence, Quantity I > Quantity II > Quantity III

12. Questions

Answer: D

Let the quantity of water in the mixture = 'x' L

The quantity of milk in the mixture = 48 L

The cost of pure milk = ₹75 per L

Since, the selling price of the mixture = ₹81 per L

And the profit percentage = 35%

So, the cost price of the mixture $= 81 * (100/135) = ₹60$ per L

Now,

$$(48 * 75)/(48 + x) = 60$$

$$48 + x = 60$$

$$x = 12$$

The quantity of water in the mixture = 12 L

Quantity I:

The difference between the quantities of milk and water in the mixture $= 48 - 12 = 36$ L

Quantity II:

The quantity of milk in the mixture = 48 L

The quantity of water in the mixture = 12 L

If the cost of pure milk were ₹50 per L:

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Then the cost of the mixture would have been = $(48 * 50)/(48 + 12) = ₹40$ per L

Quantity III:

The initial ratio of milk to water in the mixture = $48:12 = 4:1$

When $16\frac{2}{3}\%$ quantity of mixture is replaced with the same quantity of water:

The new quantity of milk in the mixture = $83\frac{1}{3}\%$ of $48 = 40$ L

Hence, Quantity I < Quantity II = Quantity III

13. Questions

Answer: C

Let the smallest number and the 2nd largest number be 'x' and '2x' respectively.

Also let the 2nd smallest number and the largest number be '5y' and '9y' respectively.

Since, the average of the four numbers = 31

$$\text{So, } x + 2x + 5y + 9y = 4 * 31$$

$$3x + 14y = 124 \text{ -----(1)}$$

Since, 150% of the smallest number is equal to the 60% of the largest number.

So,

$$x * (150/100) = 9y * (60/100)$$

$$x * 5 = 9y * 2$$

$$x = 18y/5 \text{ -----(2)}$$

From equations (1) and (2):

$$3 * (18y/5) + 14y = 124$$

$$54y + 70y = 620$$

$$y = 5$$

From equation (1):

$$x = (18 * 5)/5$$

$$x = 18$$

The smallest number = 18

The 2nd smallest number = $5 * 5 = 25$

The 2nd largest number = $2 * 18 = 36$

The largest number = $9 * 5 = 45$

Quantity I:

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The average of smallest number, the 2nd largest number and the largest number = $(18 + 36 + 45)/3 = 33$

Quantity II:

Since, the 50% of the smallest number = 50% of 18 = 9

And the 88(8/9)% of the largest number = 88(8/9)% of 45 = 40

So, the required difference = $40 - 9 = 31$

Quantity III:

So, the new number = $18 - 3 = 15$

So, the required percentage = $(15/25) * 100 = 60\%$

Hence, Quantity III > Quantity I > Quantity II

14. Questions

Answer: E

Since, the SP of article B = ₹7200

And the loss occurred on article B = $14(2/7)\% = (100/7)\%$

So, the CP of article B = $7200 * (700/600) = ₹8400$

Since, the total SP of both the articles = $2 * 7200 = ₹14400$

So, the total CP of both the articles = ₹14400

And the CP of article A = $14400 - 8400 = ₹6000$

And the value of X = $[(7200 - 6000)/6000] * 100 = 20\%$

Quantity I:

Since, the CP of article A = ₹6000

So, the MP of article A = $(100 + X + 5)\%$ of 6000 = $(100 + 20 + 5)\%$ of 6000 = ₹7500

And the SP of article A = $7500 - 900 = ₹6600$

So, the profit per cent on article A = $[(6600 - 6000)/6000] * 100 = 10\%$

Quantity II:

Since, the CP of article A = ₹6000

So, the SP of article A = 90% of 6000 = ₹5400

Since, the CP of article B = ₹8400

So, the SP of article B = 125% of 8400 = ₹10500

Since, the total profit received = $(5400 + 10500) - (6000 + 8400) = ₹1500$

So, the profit per cent = $[1500/(6000 + 8400)] * 100 = 10.45\%$ (approx.)

Quantity III:

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Since, the CP of article A = ₹6000

And the CP of article B = ₹8400

So, the required percentage = $[(8400 - 6000)/6000] * 100 = 40\%$

Hence, Quantity III > Quantity II > Quantity I

15. Questions

Answer: A

Let the speed of bike B = 'x' m/s

So, the speed of bike A = $x - 18 * (5/18) = (x - 5)$ m/s

And the speed of bike C = $(x - 5) + 10.8 * (5/18) = (x - 5) + 3 = (x - 2)$ m/s

Since, the distance covered by bike B in 4.5 minutes is 2.7 km less than that covered by boat C in 7.5 minutes.

So,

$$[(x - 2) * 7.5 * 60] - [x * 4.5 * 60] = 2.7 * 1000$$

$$450x - 900 - 270x = 2700$$

$$180x = 3600$$

$$x = 20$$

The speed of bike A = $20 - 5 = 15$ m/s

The speed of bike B = 20 m/s

The speed of bike C = $20 - 2 = 18$ m/s

Quantity I:

Since, the speed of bike A = 15 m/s

And the speed of bike C = 18 m/s

So, the time, in which both the bikes will cross each other = $825/(15 + 18) = 25$ seconds

Quantity II:

Since, the speed of bike A = 15 m/s

And the speed of bike B = 20 m/s

So, the required percentage = $[(20 - 15)/20] * 100 = 25\%$

Quantity III:

Since, the speed of bike B = 20 m/s

So, the speed of bike D = 80% of 20 = 16 m/s

And the time taken by bike D to cross 352 m long tunnel = $352/16 = 22$ seconds

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Hence, Quantity I = Quantity II > Quantity III

16. Questions

Since, the number of students in classes B and D, who were born in 2009, are prime numbers between 15 and 22.

And the only prime numbers between 15 and 22 are 17 and 19.

So, the number of students in classes B and D, who were born in 2009, will be 17 and 19, but not in the same order.

Since, the number of students in class B, who were born in 2009, is more than those in class D, who were born in 2009.

So, the number of students in class B, who were born in 2009 = 19

And the number of students in class D, who were born in 2009 = 17

Since, the number of students in class A, who were born in 2009, is 2 less than those in class D, who were born in 2009 and also 1 more than those in class E, who were born in 2009.

So, the number of students in class A, who were born in 2009 = $17 - 2 = 15$

And the number of students in class E, who were born in 2009 = $15 - 1 = 14$

Since, the number of students in class C, who were born in 2009, is 5 less than double of the number of students in class A, who were born in 2009.

So, the number of students in class C, who were born in 2009 = $(2 * 15 - 5) = 25$

Class	No. of students, who were born in 2009	No. of students, who were born in 2008	No. of students, who were born in 2007
A	15	$15 + 9 = 24$	$24 - 6 = 18$
B	19	$19 - 4 = 15$	$15 + 7 = 22$
C	25	$25 - 5 = 20$	$20 - 4 = 16$
D	17	$17 + 4 = 21$	$21 - 9 = 12$
E	14	$14 + 2 = 16$	$16 + 9 = 25$

Answer: C

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In class D:

The age of each of those students in 2022, who were born in 2007 = 15 years

The age of each of those students in 2022, who were born in 2008 = 14 years

The age of each of those students in 2022, who were born in 2009 = 13 years

So,

$$M = [(12 * 15) + (21 * 14) + (17 * 13)] / (12 + 21 + 17)$$

$$M = 695/50$$

$M = 13.9$ years

In class E:

The age of each of those students in 2020, who were born in 2008 = 12 years

The age of each of those students in 2024, who were born in 2009 = 15 years

So, the required average = $[(16 * 12) + (14 * 15)] / (16 + 14) = 13.4$ years = $(M - 0.5)$ years

17. Questions

Since, the number of students in classes B and D, who were born in 2009, are prime numbers between 15 and 22.

And the only prime numbers between 15 and 22 are 17 and 19.

So, the number of students in classes B and D, who were born in 2009, will be 17 and 19, but not in the same order.

Since, the number of students in class B, who were born in 2009, is more than those in class D, who were born in 2009.

So, the number of students in class B, who were born in 2009 = 19

And the number of students in class D, who were born in 2009 = 17

Since, the number of students in class A, who were born in 2009, is 2 less than those in class D, who were born in 2009 and also 1 more than those in class E, who were born in 2009.

So, the number of students in class A, who were born in 2009 = $17 - 2 = 15$

And the number of students in class E, who were born in 2009 = $15 - 1 = 14$

Since, the number of students in class C, who were born in 2009, is 5 less than double of the number of students in class A, who were born in 2009.

So, the number of students in class C, who were born in 2009 = $(2 * 15 - 5) = 25$

Class	No. of students, who were born in 2009	No. of students, who were born in 2008	No. of students, who were born in 2007
A	15	$15 + 9 = 24$	$24 - 6 = 18$
B	19	$19 - 4 = 15$	$15 + 7 = 22$
C	25	$25 - 5 = 20$	$20 - 4 = 16$
D	17	$17 + 4 = 21$	$21 - 9 = 12$
E	14	$14 + 2 = 16$	$16 + 9 = 25$

Answer: B

In class B:

The age of each of those students in 2021, who were born in 2007 = 14 years

The age of each of those students in 2021, who were born in 2008 = 13 years

The age of each of those students in 2021, who were born in 2009 = 12 years

Since, the number of students, who were born in 2007 = 22

The number of students, who were born in 2008 = $15 - 6 = 9$

And the number of students, who were born in 2009 = 19

So, the average age of class B in 2021 = $[(22 * 14) + (9 * 13) + (19 * 12)] / (22 + 9 + 19) = 653/50$
 = 13.06 years

18. Questions

Since, the number of students in classes B and D, who were born in 2009, are prime numbers between 15 and 22.

And the only prime numbers between 15 and 22 are 17 and 19.

So, the number of students in classes B and D, who were born in 2009, will be 17 and 19, but not in the same order.

Since, the number of students in class B, who were born in 2009, is more than those in class D, who were born in 2009.

So, the number of students in class B, who were born in 2009 = 19

And the number of students in class D, who were born in 2009 = 17

Since, the number of students in class A, who were born in 2009, is 2 less than those in class D, who were born in 2009 and also 1 more than those in class E, who were born in 2009.

So, the number of students in class A, who were born in 2009 = $17 - 2 = 15$

And the number of students in class E, who were born in 2009 = $15 - 1 = 14$

Since, the number of students in class C, who were born in 2009, is 5 less than double of the number of students in class A, who were born in 2009.

So, the number of students in class C, who were born in 2009 = $(2 * 15 - 5) = 25$

Class	No. of students, who were born in 2009	No. of students, who were born in 2008	No. of students, who were born in 2007
A	15	$15 + 9 = 24$	$24 - 6 = 18$
B	19	$19 - 4 = 15$	$15 + 7 = 22$
C	25	$25 - 5 = 20$	$20 - 4 = 16$
D	17	$17 + 4 = 21$	$21 - 9 = 12$
E	14	$14 + 2 = 16$	$16 + 9 = 25$

Answer: E

In class E:

Since, the number of students, who were born in 2007 = 25

So, the number of girls, who were born in 2007 = $25 * (2/5) = 10$

Since, the number of students, who were born in 2008 = 16

So, the number of girls, who were born in 2008 = $16 * (5/8) = 10$

Since, the number of students, who were born in 2009 = 14

So, the number of girls, who were born in 2009 = $14 * (5/7) = 10$

Required percentage = $[(10 + 10 + 10)/(25 + 16 + 14)] * 100 = 54(6/11)\%$

19. Questions

Since, the number of students in classes B and D, who were born in 2009, are prime numbers between 15 and 22.

And the only prime numbers between 15 and 22 are 17 and 19.

So, the number of students in classes B and D, who were born in 2009, will be 17 and 19, but not in the same order.

Since, the number of students in class B, who were born in 2009, is more than those in class D, who were born in 2009.

So, the number of students in class B, who were born in 2009 = 19

And the number of students in class D, who were born in 2009 = 17

Since, the number of students in class A, who were born in 2009, is 2 less than those in class D, who were born in 2009 and also 1 more than those in class E, who were born in 2009.

So, the number of students in class A, who were born in 2009 = $17 - 2 = 15$

And the number of students in class E, who were born in 2009 = $15 - 1 = 14$

Since, the number of students in class C, who were born in 2009, is 5 less than double of the number of students in class A, who were born in 2009.

So, the number of students in class C, who were born in 2009 = $(2 * 15 - 5) = 25$

Class	No. of students, who were born in 2009	No. of students, who were born in 2008	No. of students, who were born in 2007
A	15	$15 + 9 = 24$	$24 - 6 = 18$
B	19	$19 - 4 = 15$	$15 + 7 = 22$
C	25	$25 - 5 = 20$	$20 - 4 = 16$
D	17	$17 + 4 = 21$	$21 - 9 = 12$
E	14	$14 + 2 = 16$	$16 + 9 = 25$

Answer: A

Since, the age of each of those students of class A in 2022, who were born in 2007 = 15 years

And the age of each of those students of class D in 2017, who were born in 2007 = 10 years

$$\text{So, } P = [(18 * 15) + (12 * 10)] / (18 + 12)$$

$$P = 13 \text{ years}$$

Since, the age of each of those students of class A in 2025, who were born in 2009 = 16 years

And the age of each of those students of class C in 2025, who were born in 2009 = 16 years

$$\text{So, } Q = [(15 * 16) + (25 * 16)] / (15 + 25)$$

$$Q = 16 \text{ years}$$

From X:

Since, 13 and 16 are co-prime with each other.

So, X is true.

From Y:

$$\text{Value of } (Q^2 - P^2) = (16^2 - 13^2) = (256 - 169) = 87$$

Since, 87 is not a prime number.

So, Y is not true.

From Z:

$$\text{Value of } (P^2 + Q^2) = (13^2 + 16^2) = (169 + 256) = 425$$

Since, 425 is divisible by 25.

So, Z is true.

Hence, only X and Z are true.

20. Questions

Since, the number of students in classes B and D, who were born in 2009, are prime numbers between 15 and 22.

And the only prime numbers between 15 and 22 are 17 and 19.

So, the number of students in classes B and D, who were born in 2009, will be 17 and 19, but not in the same order.

Since, the number of students in class B, who were born in 2009, is more than those in class D, who were born in 2009.

$$\text{So, the number of students in class B, who were born in 2009} = 19$$

$$\text{And the number of students in class D, who were born in 2009} = 17$$

Since, the number of students in class A, who were born in 2009, is 2 less than those in class D, who were born in 2009 and also 1 more than those in class E, who were born in 2009.

$$\text{So, the number of students in class A, who were born in 2009} = 17 - 2 = 15$$

$$\text{And the number of students in class E, who were born in 2009} = 15 - 1 = 14$$

Since, the number of students in class C, who were born in 2009, is 5 less than double of the number of students in class A, who were born in 2009.

So, the number of students in class C, who were born in 2009 = $(2 * 15 - 5) = 25$

Class	No. of students, who were born in 2009	No. of students, who were born in 2008	No. of students, who were born in 2007
A	15	$15 + 9 = 24$	$24 - 6 = 18$
B	19	$19 - 4 = 15$	$15 + 7 = 22$
C	25	$25 - 5 = 20$	$20 - 4 = 16$
D	17	$17 + 4 = 21$	$21 - 9 = 12$
E	14	$14 + 2 = 16$	$16 + 9 = 25$

Answer: B

In class C:

The number of students, who were born in 2007 = 16

So, the new number of students, who were born in 2008 = $[100 - 10(Q - P)]\%$ of 20 = $[100 - 10(16 - 13)]\%$ of 20 = 14

The number of students, who were born in 2009 = 25

So, the total number of students in class C now = $16 + 14 + 25 = 55$

@QuizByMickey